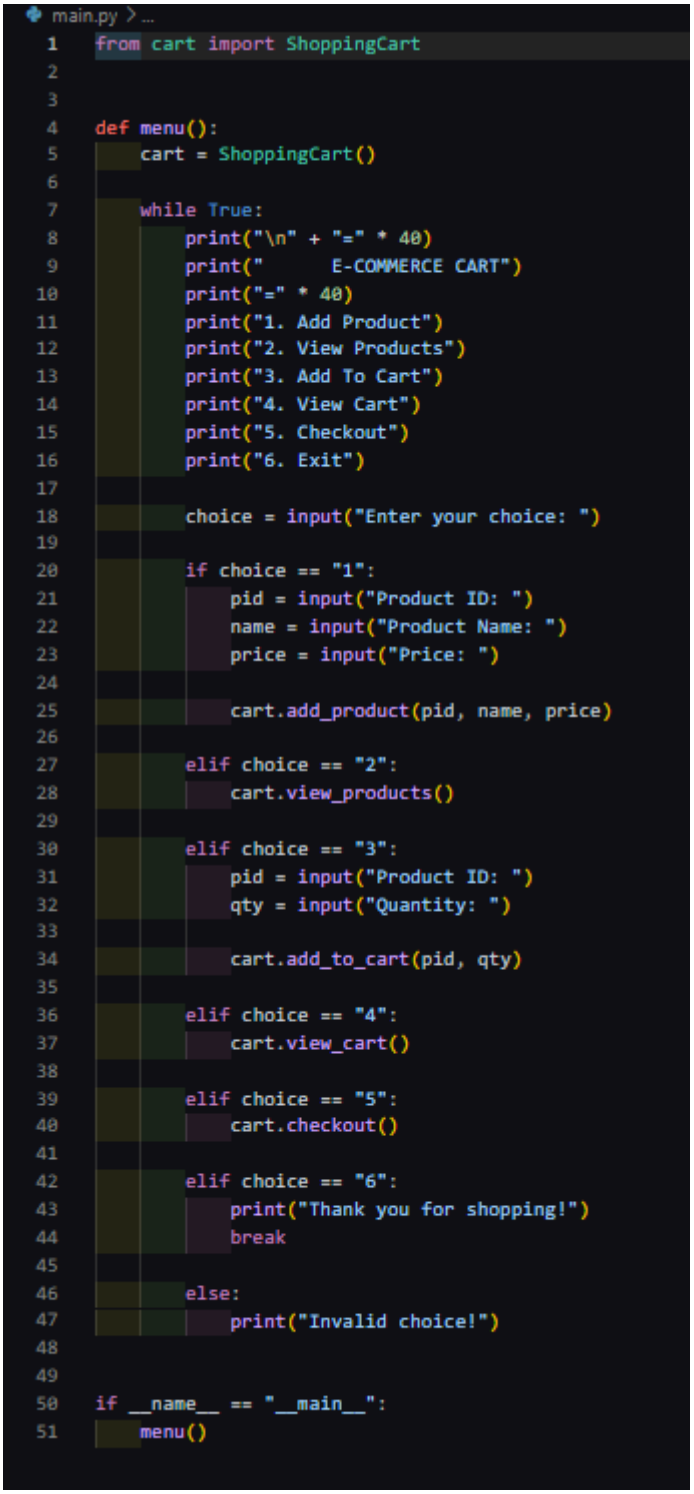
pycache

cart.py

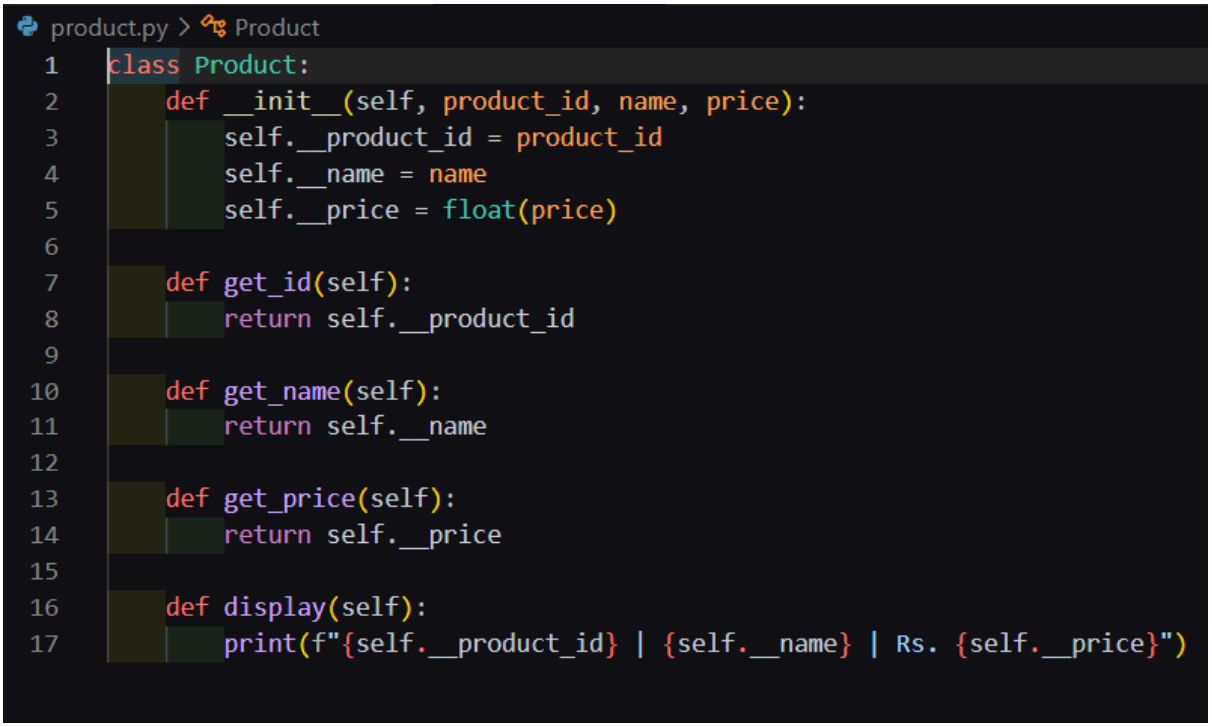
```python
from product import Product


class ShoppingCart:
    def __init__(self):
        self.__products = []
        self.__cart = []

    def add_product(self, pid, name, price):
        self.__products.append(Product(pid, name, price))
        print("Product added successfully!")

    def view_products(self):
        if not self.__products:
            print("No products available.")
            return

        print("\n===== PRODUCTS =====")

        for product in self.__products:
            product.display()

    def add_to_cart(self, pid, qty):
        for product in self.__products:
            if product.get_id() == pid:
                self.__cart.append((product, int(qty)))
                print("Added to cart!")
                return

        print("Product not found!")

    def view_cart(self):
        if not self.__cart:
            print("Cart is empty.")
            return

        subtotal = 0

        print("\n===== SHOPPING CART =====")

        for product, qty in self.__cart:
            total = product.get_price() * qty
            subtotal += total
            print(f"{product.get_name()} x{qty} = Rs. {total}")

        discount = subtotal * 0.10 if subtotal >= 5000 else 0
        vat = (subtotal - discount) * 0.13
        grand_total = subtotal - discount + vat

        print("---------------------------")
        print(f"Subtotal : Rs. {subtotal}")
        print(f"Discount : Rs. {discount}")
        print(f"VAT      : Rs. {vat}")
        print(f"Total    : Rs. {grand_total}")

    def checkout(self):
        if not self.__cart:
            print("Cart is empty.")
            return

        self.view_cart()
        print("\nCheckout Successful!")
        self.__cart.clear()
```

main.py



```python
from cart import ShoppingCart


def menu():
    cart = ShoppingCart()

    while True:
        print("\n" + "=" * 40)
        print("      E-COMMERCE CART")
        print("=" * 40)
        print("1. Add Product")
        print("2. View Products")
        print("3. Add To Cart")
        print("4. View Cart")
        print("5. Checkout")
        print("6. Exit")

        choice = input("Enter your choice: ")

        if choice == "1":
            pid = input("Product ID: ")
            name = input("Product Name: ")
            price = input("Price: ")

            cart.add_product(pid, name, price)

        elif choice == "2":
            cart.view_products()

        elif choice == "3":
            pid = input("Product ID: ")
            qty = input("Quantity: ")

            cart.add_to_cart(pid, qty)

        elif choice == "4":
            cart.view_cart()

        elif choice == "5":
            cart.checkout()

        elif choice == "6":
            print("Thank you for shopping!")
            break

        else:
            print("Invalid choice!")


if __name__ == "__main__":
    menu()
```

product.py



File 1: `product.py`

Purpose

The `product.py` file defines the **Product** class, which stores information about each product, including its ID, name, and price.

Functions

- `__init__()` – Initializes the product ID, name, and price.
- `get_id()` – Returns the product ID.
- `get_name()` – Returns the product name.
- `get_price()` – Returns the product price.
- `display()` – Displays the product details in a readable format.

Summary

This file acts as the product model and provides the necessary methods to access product information safely.

File 2: `cart.py`

Purpose

The `cart.py` file contains the **ShoppingCart** class, which manages products, shopping cart operations, billing, and checkout.

Functions

- **`add_product()`** – Adds a new product to the product list.
- **`view_products()`** – Displays all available products.
- **`add_to_cart()`** – Adds a selected product and quantity to the cart.
- **`view_cart()`** – Calculates the subtotal, discount, VAT, and total bill.
- **`checkout()`** – Displays the final bill and clears the cart after purchase.

Summary

This file performs the main operations of the shopping cart system, including managing products and calculating the final payment.

File 3: `main.py`

Purpose

The `main.py` file is the main program that provides a menu-driven interface for users to interact with the shopping cart system.

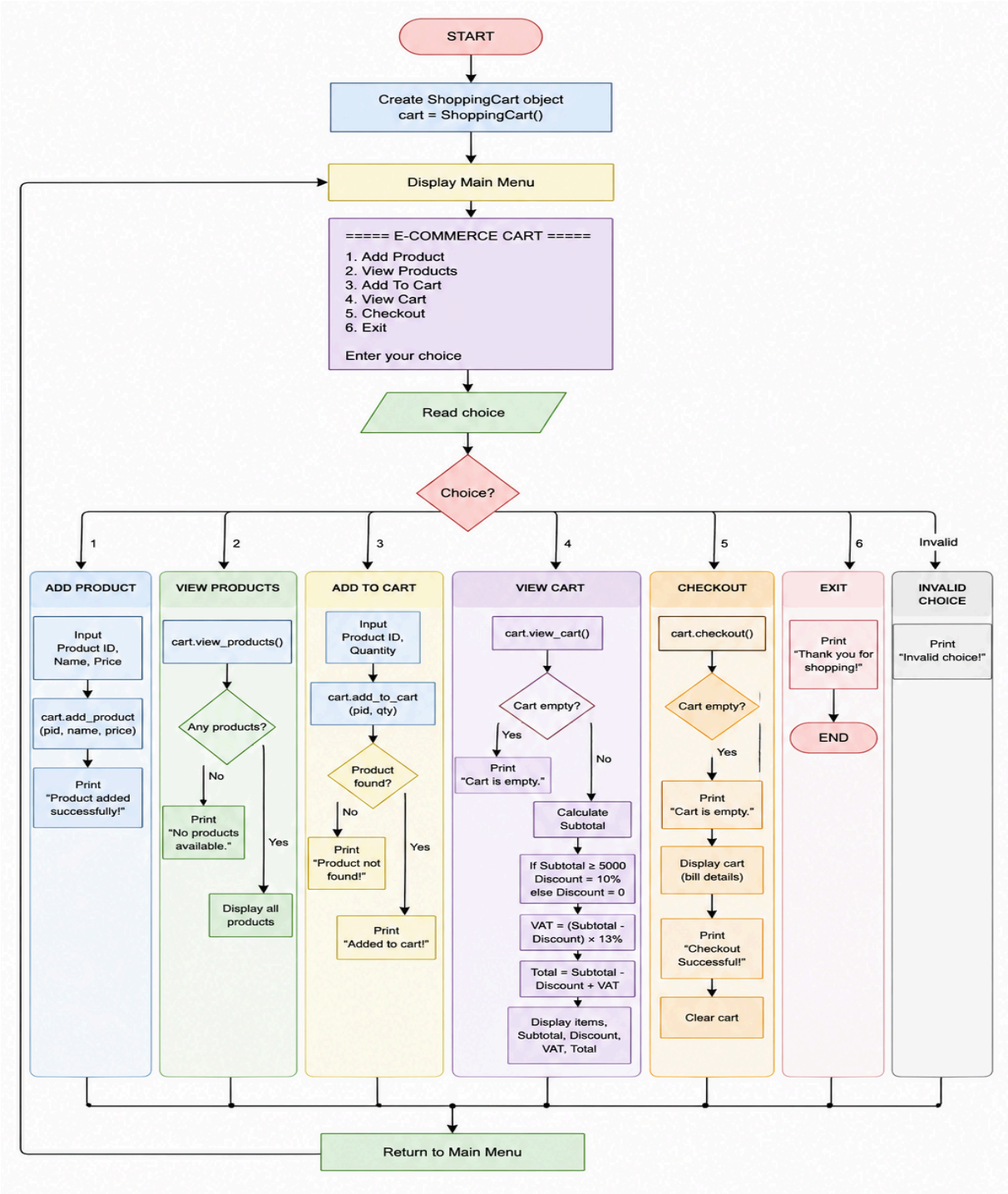
Working

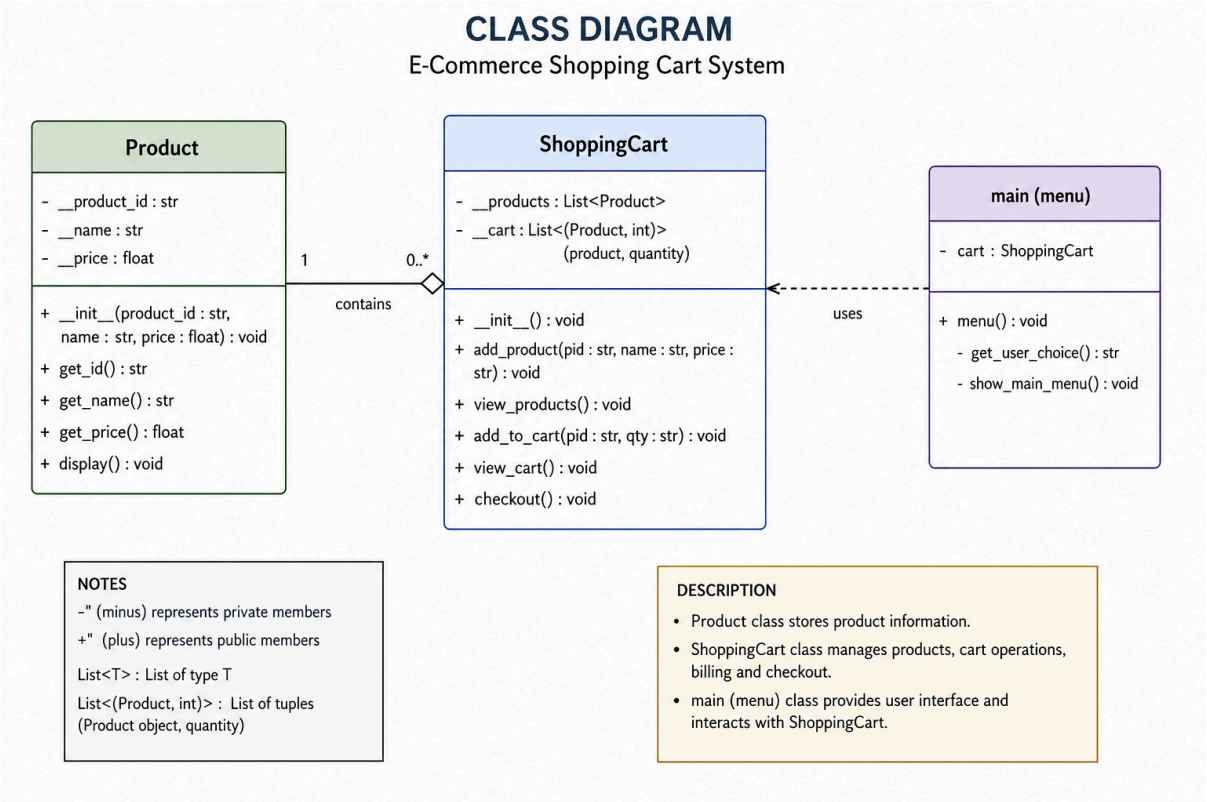
- Creates a `ShoppingCart` object.
- Displays menu options to the user.
- Accepts user input and performs actions such as adding products, viewing products, adding items to the cart, viewing the cart, checking out, or exiting the program.
- Calls the appropriate methods from the `ShoppingCart` class based on the user's choice.

Summary

This file controls the execution of the program and connects the user with the shopping cart functionalities through a simple menu interface.

FLOWCHART:


START
Create ShoppingCart object
cart = ShoppingCart()
Display Main Menu
===== E-COMMERCE CART =====
1. Add Product
2. View Products
3. Add To Cart
4. View Cart
5. Checkout
6. Exit
Enter your choice
Read choice
Choice?
1
2
3
4
5
6
Invalid
ADD PRODUCT
Input Product ID, Name, Price
cart.add_product (pid, name, price)
Print "Product added successfully!"
VIEW PRODUCTS
cart.view_products()
Any products?
No
Print "No products available."
Yes
Display all products
ADD TO CART
Input Product ID, Quantity
cart.add_to_cart (pid, qty)
Product found?
No
Print "Product not found!"
Yes
Print "Added to cart!"
VIEW CART
cart.view_cart()
Cart empty?
Yes
Print "Cart is empty."
No
Calculate Subtotal
If Subtotal ≥ 5000 Discount = 10% else Discount = 0
VAT = (Subtotal - Discount) × 13%
Total = Subtotal - Discount + VAT
Display items, Subtotal, Discount, VAT, Total
CHECKOUT
cart.checkout()
Cart empty?
Yes
Print "Cart is empty."
Display cart (bill details)
Print "Checkout Successful!"
Clear cart
EXIT
Print "Thank you for shopping!"
END
INVALID CHOICE
Print "Invalid choice!"
Return to Main Menu



`

File Structure

E-Commerce Shopping Cart System
│
├── main.py → Main program (Menu)
├── cart.py → Shopping cart operations
├── product.py → Product class
└── __pycache__/ → Compiled Python files

Conclusion

The E-Commerce Shopping Cart System is a simple Python application developed using object-oriented programming principles. It is divided into three main files: `product.py` stores product information, `cart.py` manages shopping cart operations and billing, and `main.py` provides a menu-driven interface for user interaction. The program allows users to add products, view available products, add items to the cart, calculate discounts and VAT, and complete the checkout process. This modular structure makes the system easy to understand, maintain, and extend with additional features in the future.